

4. 设 x 不在 p 中自由出现. 求证:

$$1^\circ \vdash (p \rightarrow \forall x q) \rightarrow \forall x (p \rightarrow q).$$

$$2^\circ \vdash (p \rightarrow \exists x q) \rightarrow \exists x (p \rightarrow q).$$

证明:

$$1^\circ \text{ 先证 } \{p \rightarrow \forall x q\} \vdash \forall x (p \rightarrow q)$$

$$(1) \quad p \rightarrow \forall x q \quad \text{假设}$$

$$(2) \quad \forall x q \rightarrow q \quad (K4)$$

$$(3) \quad (p \rightarrow \forall x q) \rightarrow ((\forall x q \rightarrow q) \rightarrow (p \rightarrow q)) \quad \text{假设} \equiv \text{结论}$$

$$(4) \quad (\forall x q \rightarrow q) \rightarrow (p \rightarrow q) \quad (1), (3), \text{MP}$$

$$(5) \quad p \rightarrow q \quad (2), (4), \text{MP}$$

$$(6) \quad \forall x (p \rightarrow q) \quad (5), \text{Gen}$$

又 x 在 $\{p \rightarrow \forall x q\}$ 中不自由出现, 故由演绎定理:

$$\vdash (p \rightarrow \forall x q) \rightarrow \forall x (p \rightarrow q) \quad \text{证毕}$$

$$2^\circ \text{ 先证 } \{p \rightarrow \exists x q\} \vdash \exists x (p \rightarrow q)$$

利用)和律, 证明 $\{p \rightarrow \exists x q, \forall x \neg (p \rightarrow q)\} \vdash r$ 及 $\neg r$.

$$(1) \quad p \rightarrow \neg \forall x \neg (p \rightarrow q) \quad \text{假设}$$

$$(2) \quad \forall x \neg (p \rightarrow q) \quad \text{假设}$$

$$(3) \quad \forall x \neg (p \rightarrow q) \rightarrow \neg (p \rightarrow q) \quad (K4)$$

$$(4) \quad \neg (p \rightarrow q) \quad (2), (3), \text{MP}$$

$$(5) \quad \neg (p \rightarrow q) \rightarrow p \quad \text{永真式}$$

$$(6) \quad p \quad (4), (5), \text{MP}$$

$$(7) \quad \neg \forall x \neg (p \rightarrow q) \quad (1), (6), \text{MP}$$

$$(8) \neg(p \rightarrow q) \rightarrow \neg q$$

永真式

$$(9) \neg q$$

(4), (8), mp

$$(10) \forall x \neg q$$

(9), Gen

由(7)(10)可知满足归零律. 即得证 $\{p \rightarrow \exists x q\} \vdash \exists x(p \wedge q)$

又 x 在 $\{p \rightarrow \exists x q\}$ 中不自由出现, 故由演绎定理:

$$\vdash (p \rightarrow \exists x q) \rightarrow (\exists x (p \rightarrow q)) \quad \text{证毕}$$

$k > 0$ 时, 设 $u(x) = f_i^n(t_1(x), \dots, t_n(x))$, 其中 $t_1(x), \dots, t_n(x)$ 是较低层次的项.

这时

$u(x)$

中:

$$u(t) = f_i^n(t_1(t), \dots, t_n(t)).$$

我们有

$$\begin{aligned} \varphi'(u(x)) &= \varphi'(f_i^n(t_1(x), \dots, t_n(x))) \\ &= \overline{f_i^n}(\varphi'(t_1(x)), \dots, \varphi'(t_n(x))) && (\varphi' \text{ 是项解释}) \\ &= \overline{f_i^n}(\varphi(t_1(t)), \dots, \varphi(t_n(t))) && (\text{由归纳假设}) \\ &= \varphi(f_i^n(t_1(t), \dots, t_n(t))) && (\varphi \text{ 是项解释}) \\ &= \varphi(u(t)). \end{aligned}$$

至此, 证明 1° 的归纳过程完成.

2° 对公式 $p(x)$ 在 $K(Y)$ 中的层数 k 归纳.

$$(3) \quad u(x) = x, \text{ 此时 } u(t) = t, \text{ 由 } \varphi'(x) = \varphi(t) \text{ 知 } \varphi'(u(x)) = \varphi(u(t))$$

$k > 0$ 时, 设 $u(x) = f_i^n(t_1(x), \dots, t_n(x))$, 其中 $t_1(x), \dots, t_n(x)$ 是较低层次的项. 这时: $u(t) = f_i^n(t_1(t), \dots, t_n(t))$

$$\text{有: } \varphi'(u(x)) = \varphi'(f_i^n(t_1(x), \dots, t_n(x)))$$

$$= \overline{f_i^n}(\varphi'(t_1(x)), \dots, \varphi'(t_n(x))) \quad (\varphi' \text{ 是项解释})$$

$$= \overline{f_i^n}(\varphi(t_1(t)), \dots, \varphi(t_n(t))) \quad (\text{归纳假设})$$

$$= \varphi(f_i^n(t_1(t), \dots, t_n(t))) \quad (\varphi \text{ 是项解释})$$

$$= \varphi(u(t)) \quad \text{归纳完毕. 证毕.}$$

1. 设 K 中的 $C = \{c_i\}$, $F = \{f_1^1, f_1^2, f_2^2\}$, $R = \{R_1^2\}$. 它的一个解释域是 $N = \{0, 1, 2, \dots\}$, $\overline{c_1} = 0$, $\overline{f_1^1}$ 是后继函数, $\overline{f_1^2}$ 是 $+$, $\overline{f_2^2}$ 是 $\times$, $\overline{R_1^2}$ 是 $=$. 试对以下公式分别找出 $\varphi, \psi \in \mathcal{F}_N$, 使 $|p|(\varphi) = 1$, $|p|(\psi) = 0$, 其中 p 为:

$$1^\circ R_1^2(f_1^2(x_1, x_1), f_2^2(x_2, x_3)).$$

$$2^\circ R_1^2(f_1^2(x_1, c_1), x_2) \rightarrow R_1^2(f_1^2(x_1, x_2), x_3).$$

$$3^\circ \neg R_1^2(f_2^2(x_1, x_2), f_2^2(x_2, x_3)).$$

$$4^\circ \forall x_1 R_1^2(f_2^2(x_1, x_2), x_3).$$

$$5^\circ \forall x_1 R_1^2(f_2^2(x_1, c_1), x_1) \rightarrow R_1^2(x_1, x_2).$$

1°

$$\varphi: \varphi(x_1) = \varphi(x_2) = 1 \quad \varphi(x_3) = 2$$

$$\psi: \psi(x_1) = \psi(x_2) = \psi(x_3) = 1$$

3°

$$\varphi: \varphi(x_1) = \varphi(x_2) = 1 \quad \varphi(x_3) = 2$$

$$\psi: \psi(x_1) = \psi(x_2) = \psi(x_3) = 1$$

$$5^\circ \forall x_1, (x_1 \times 0 = x_1) \rightarrow x_1 = x_2$$

$$\varphi: \varphi \text{ 的任一 } x_1 \text{ 变通 } \varphi' \text{ 都有 } \varphi'(x_1) = \varphi'(x_2)$$

$$\psi: \text{存在 } \psi \text{ 的 } x_1 \text{ 变通 } \psi', \text{ 使得 } \psi'(x_1) = 0, \text{ 且 } \psi'(x_2) \neq 0$$